Identify the Attackers' Location from Microsoft Sentinel by Using an Azure Virtual Machine as a Honeypot

By

Panpatchara Naneplai

Work Model

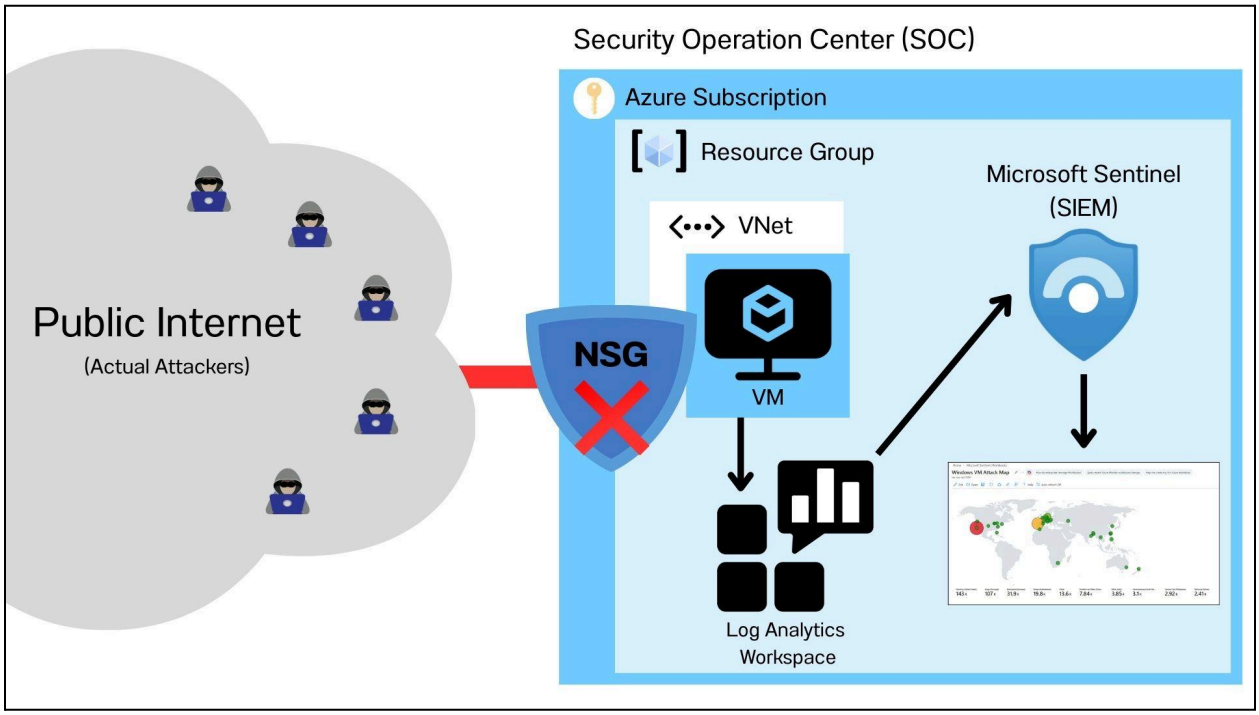


Figure 1.

This work consists of 5 parts:

1. Set up Azure Subscription
2. Create the Honey Pot (Azure Virtual Machine)
3. Log Forwarding and KQL
4. Log Enrichment and Finding Location Data
5. Attack Map Creation

Part 1: Set up Azure Subscription

Create a paid Azure subscription to access Azure products such as Virtual Machines, Virtual Networks, Microsoft Sentinel, etc.

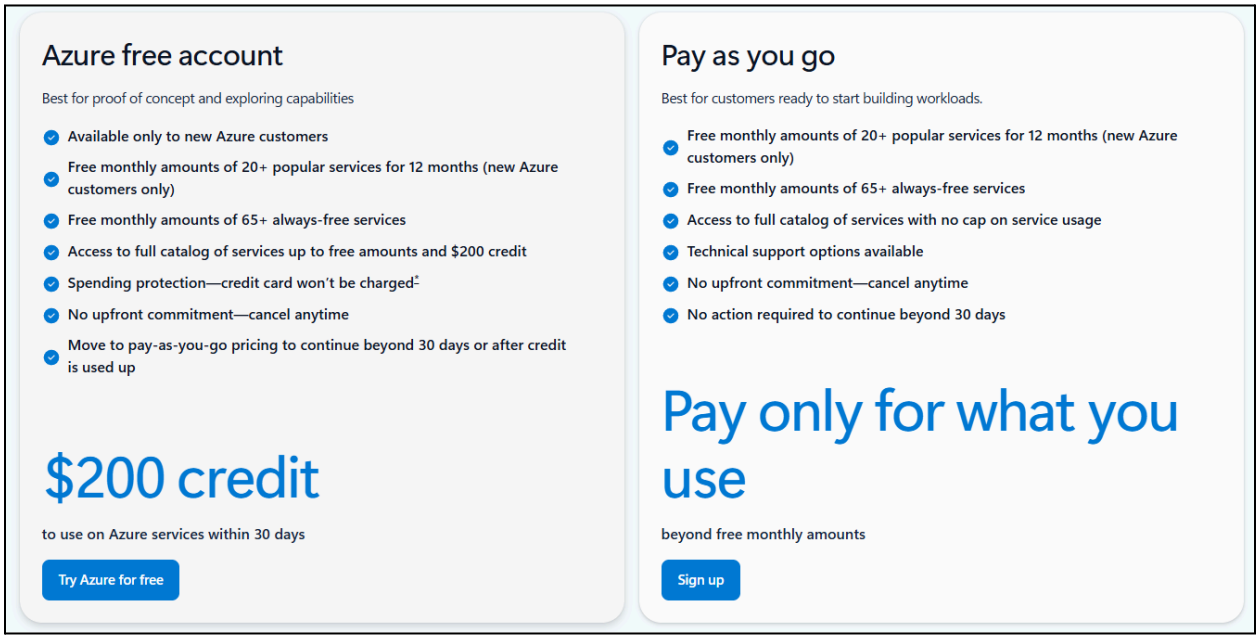


Figure 2.

As for this work, I will create an Azure free account.

Part 2: Create the Honey Pot (Azure Virtual Machine)

Part 2.1: Create a Resource Group.

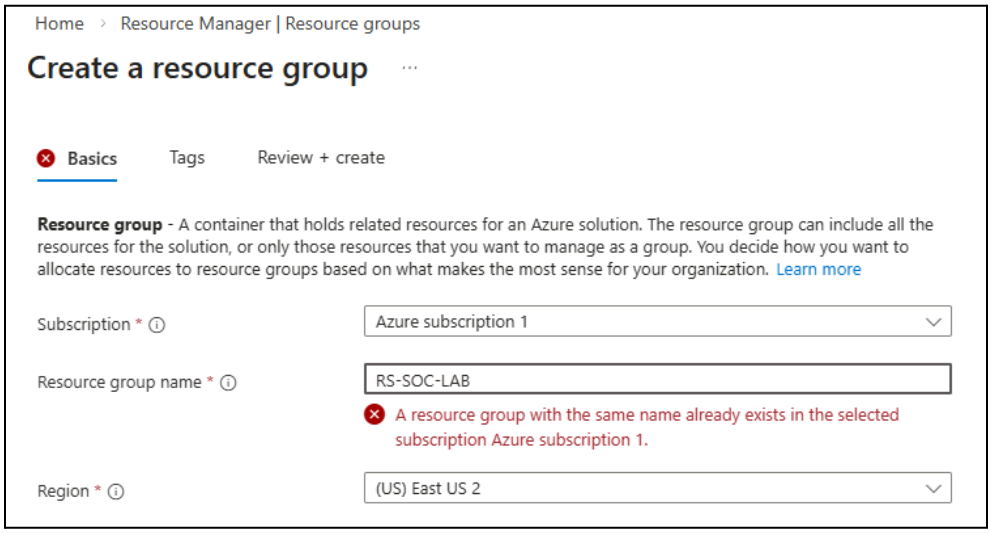


Figure 3.

Part 2.2: Create a Virtual Network and choose the resource group that was created.

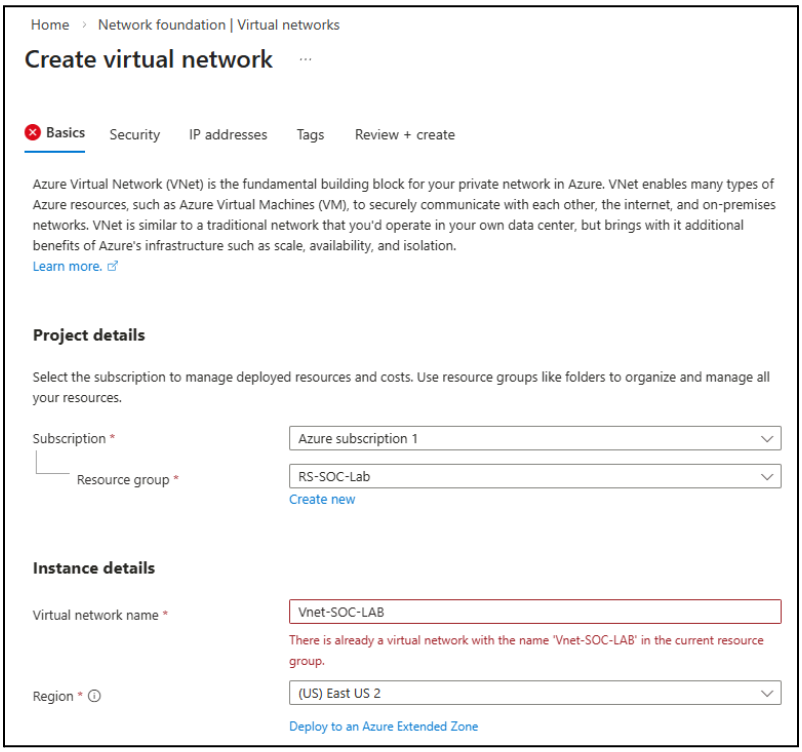


Figure 4.

Part 2.3: Create a Windows 10 Virtual Machine.
I chose the resource group that I created and the image as Windows 10 Enterprise, and chose a Virtual Network that I recently created with the subnet as the default.

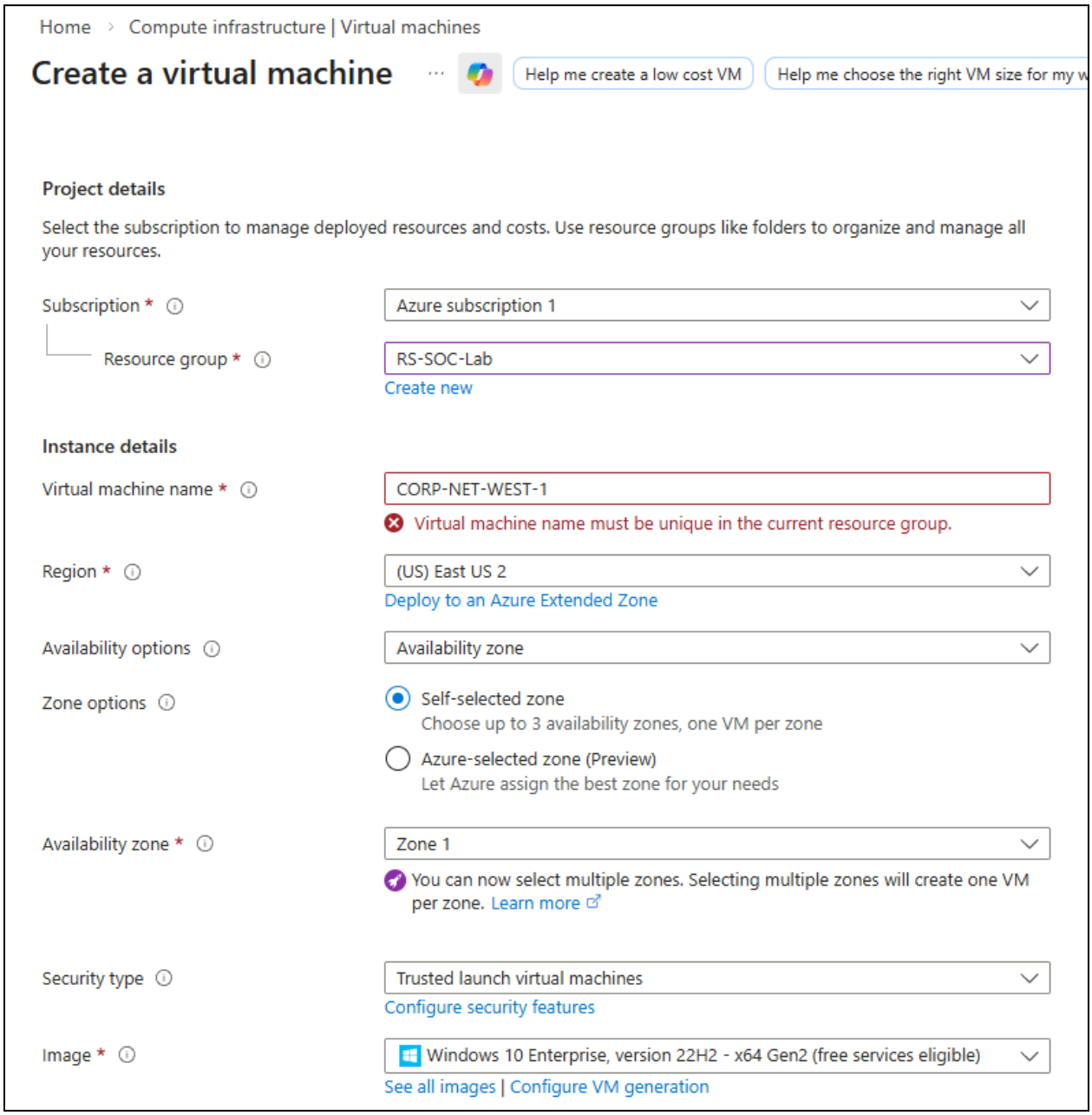


Figure 5.1

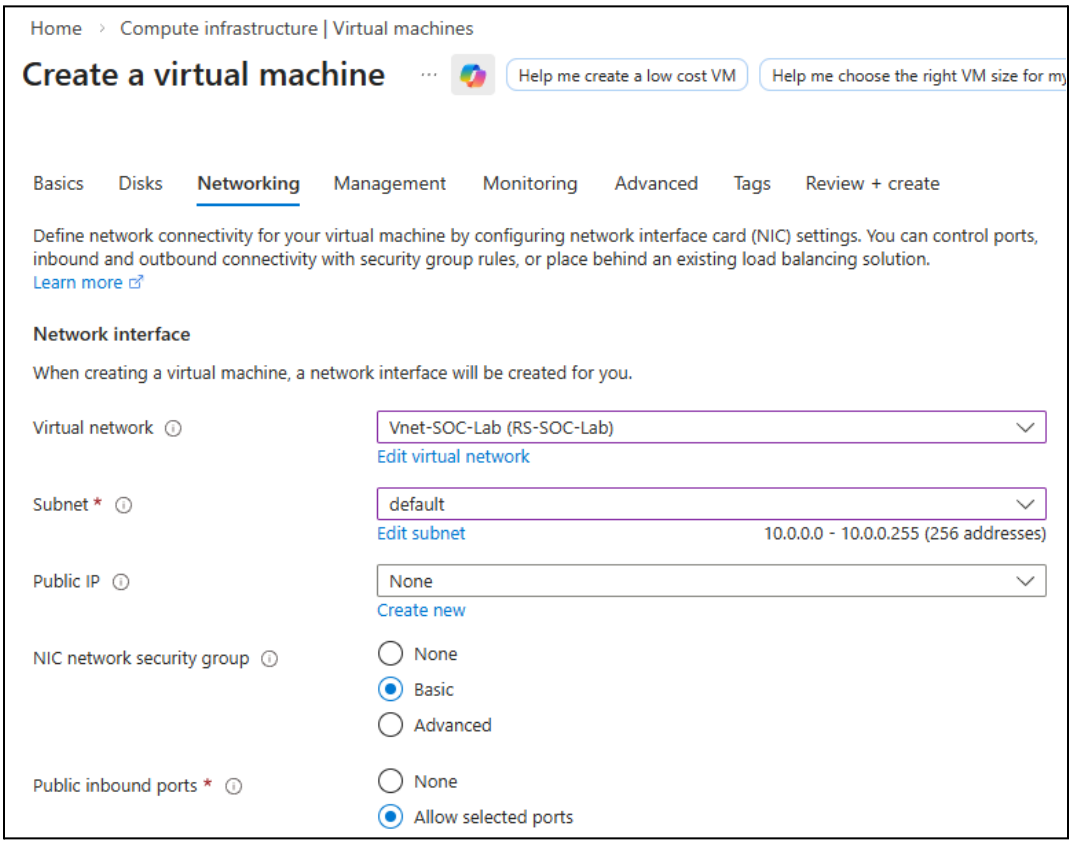

Home
Compute infrastructure | Virtual machines
Create a virtual machine
Help me create a low cost VM
Help me choose the right VM size for my
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more
Network interface
When creating a virtual machine, a network interface will be created for you.
Virtual network
Vnet-SOC-Lab (RS-SOC-Lab)
Edit virtual network
Subnet *
default
Edit subnet
10.0.0.0 - 10.0.0.255 (256 addresses)
Public IP
None
Create new
NIC network security group
None
Basic
Advanced
Public inbound ports *
None
Allow selected ports

Figure 5.2

Part 2.4: Set up an inbound security rule of the Virtual Machine to control the traffic that can enter the Virtual Network from the public network, so that we can let attackers try to get into the Virtual Machine. Then log in to the Virtual Machine via Remote Desktop Connection and turn off the firewall of the Virtual Machine.

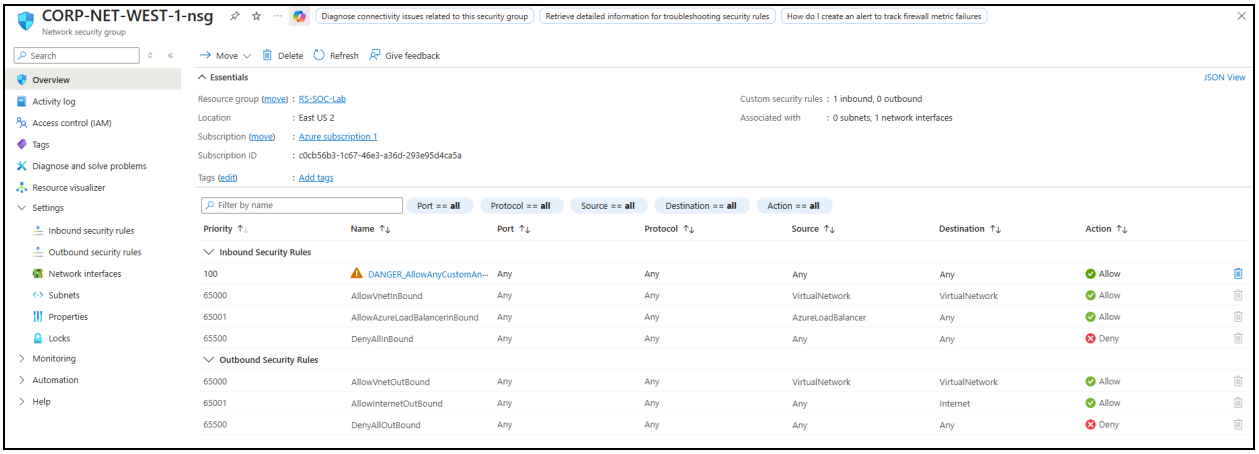


Figure 6.1

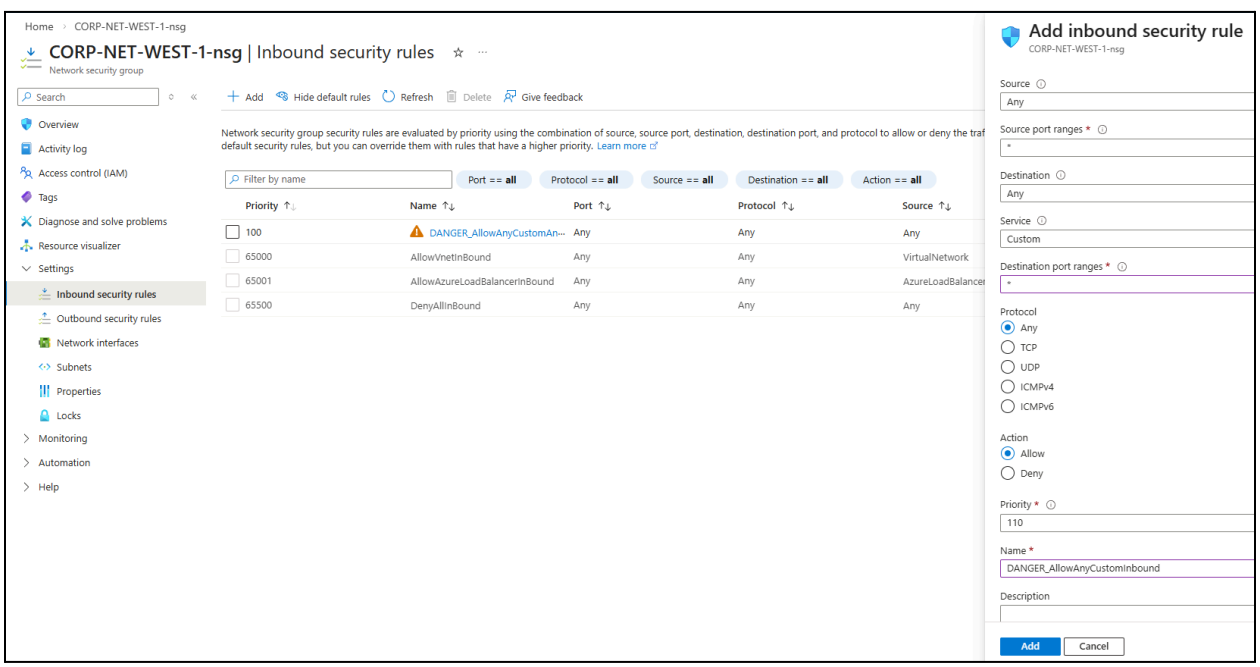


Figure 6.2

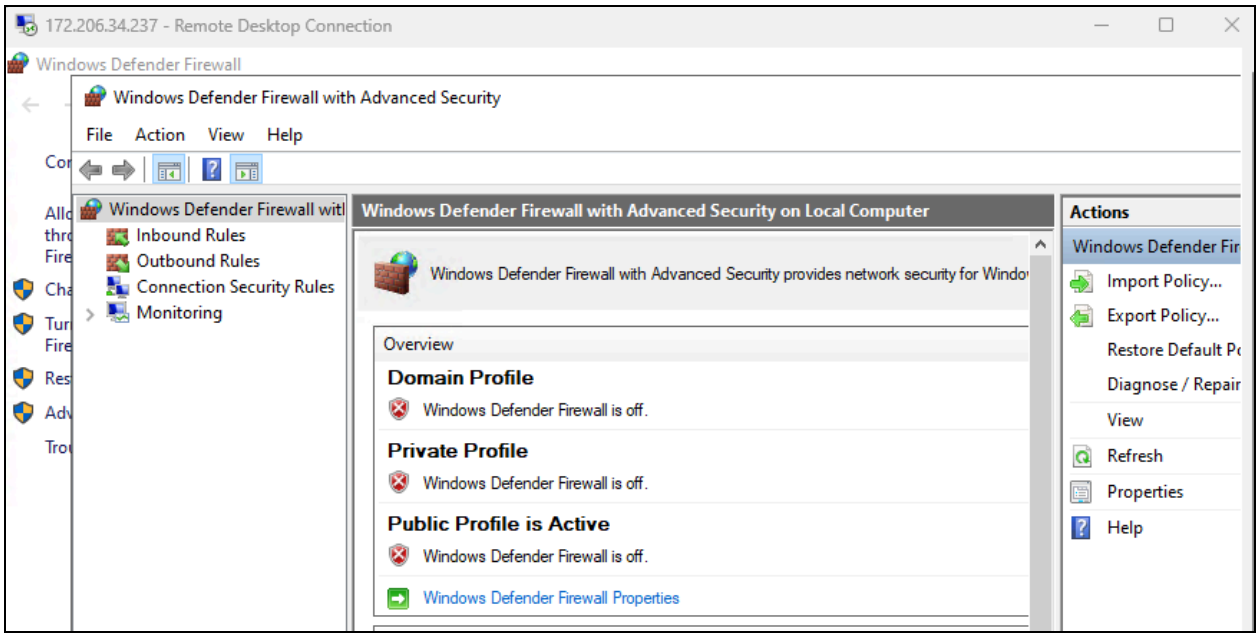

172.206.34.237 - Remote Desktop Connection
Windows Defender Firewall
Windows Defender Firewall with Advanced Security
File
Action
View
Help
Windows Defender Firewall with Advanced Security on Local Computer
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
Windows Defender Firewall with Advanced Security provides network security for Windo
Overview
Domain Profile
Windows Defender Firewall is off.
Private Profile
Windows Defender Firewall is off.
Public Profile is Active
Windows Defender Firewall is off.
Windows Defender Firewall Properties
Actions
Import Policy...
Export Policy...
Diagnose / Repair
View
Refresh
Properties
Help

Figure 6.3

Part 3: Log Forwarding and KQL

Create Log Analytics Workspace

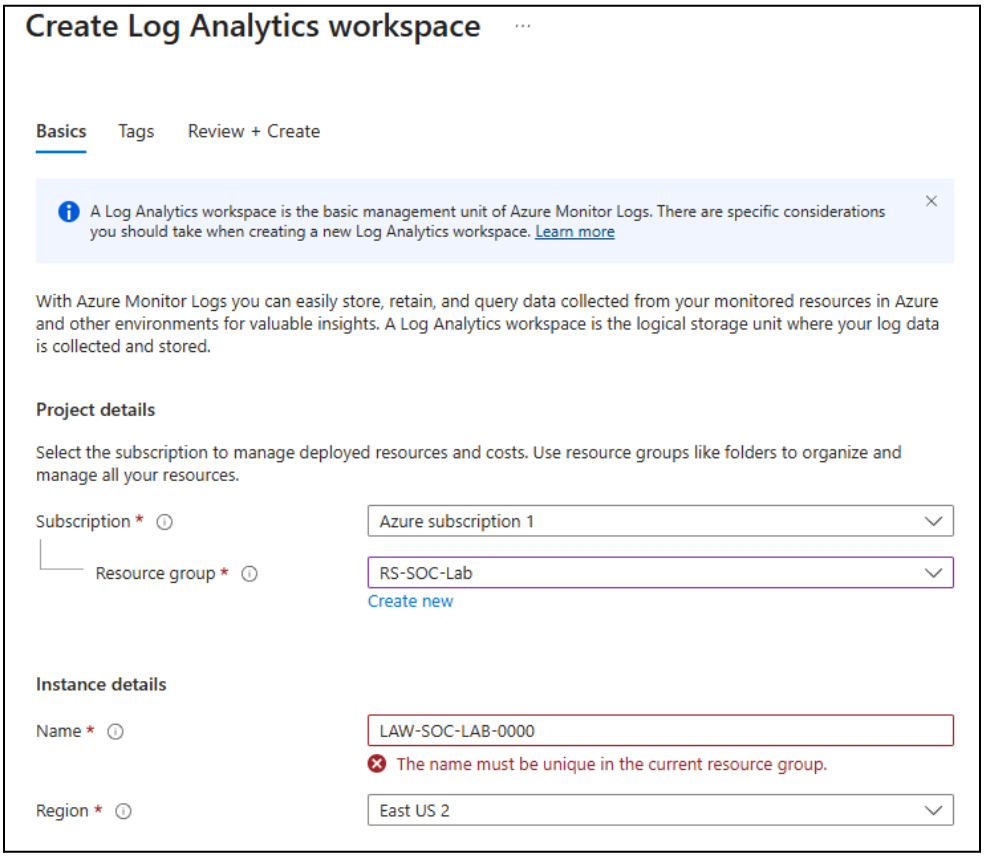


Figure 7.

Install Windows Security Event from the Microsoft Content Hub, then configure the Windows Security Event via AMA (Azure Monitor Agent) in the Data Connectors page, and create a Data Collection Rule.

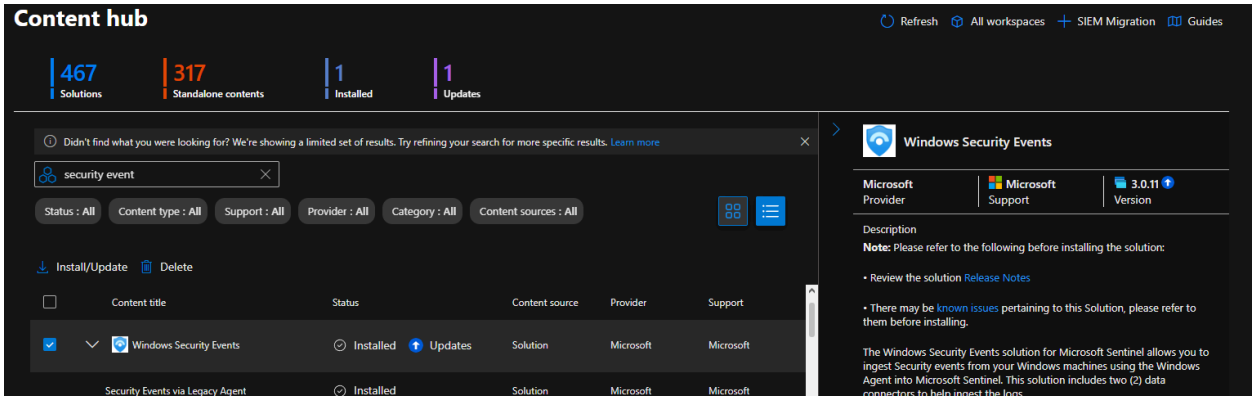


Figure 8.1

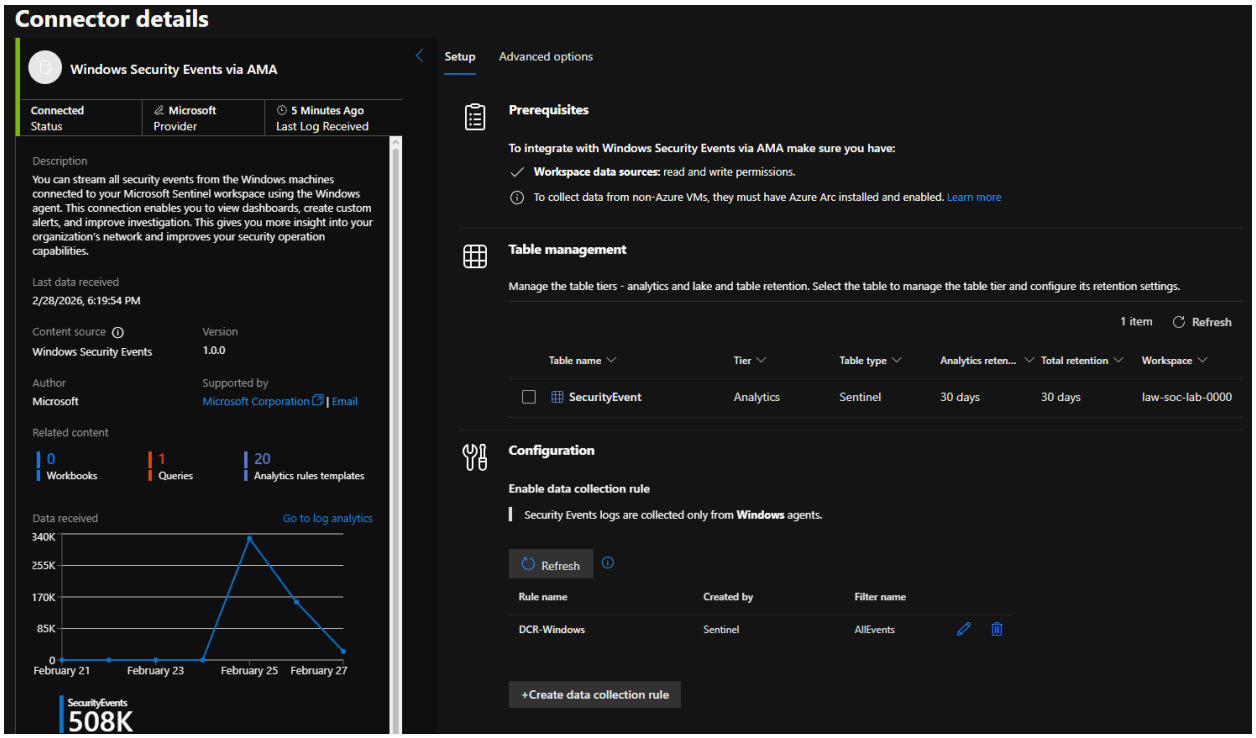


Figure 8.2

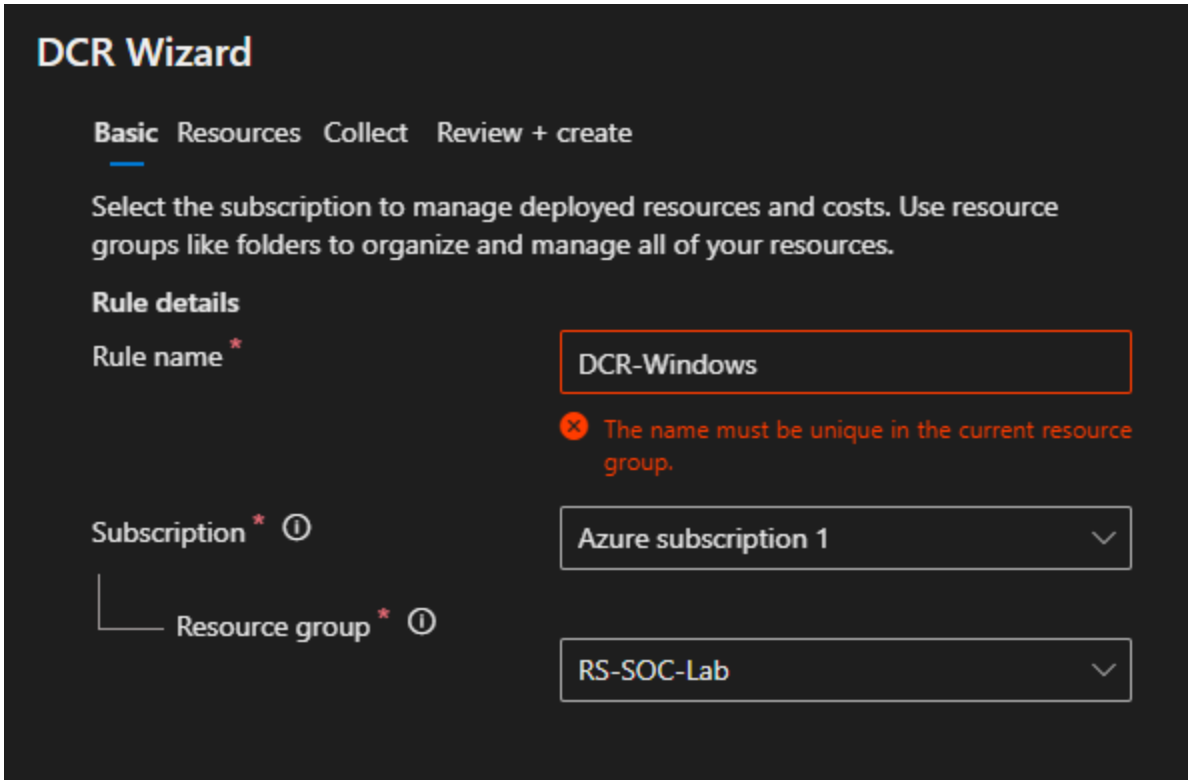


Figure 8.3

We can observe the events happening using a KQL command *'SecurityEvent'*.

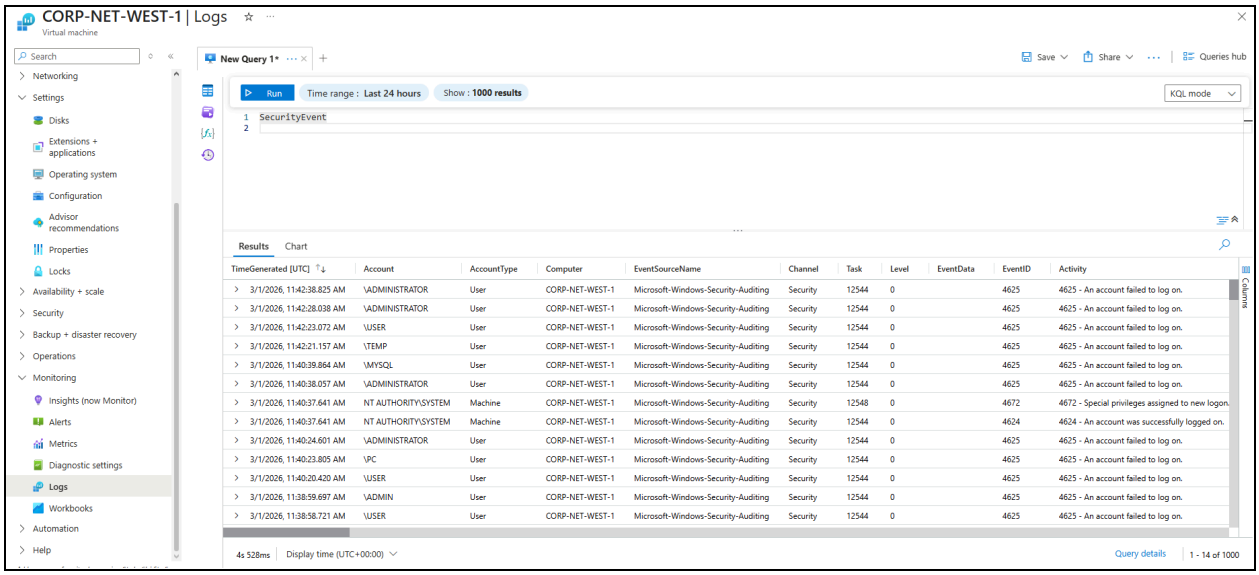


Figure 9.

Part 4: Log Enrichment and Finding Location Data

After observing the SecurityEvent logs in the Log Analytics Workspace, there is no location data, only an IP address, which we can use to derive the location data.

We are going to import a spreadsheet (as a "Sentinel Watchlist") which contains geographic information for each block of IP addresses.

We will use geoip-summarized.csv to add to the Sentinel Watchlist.

Within Sentinel, create the watchlist:

Name/Alias: geoip
Source type: Local File
Number of lines before row: 0
Search Key: network

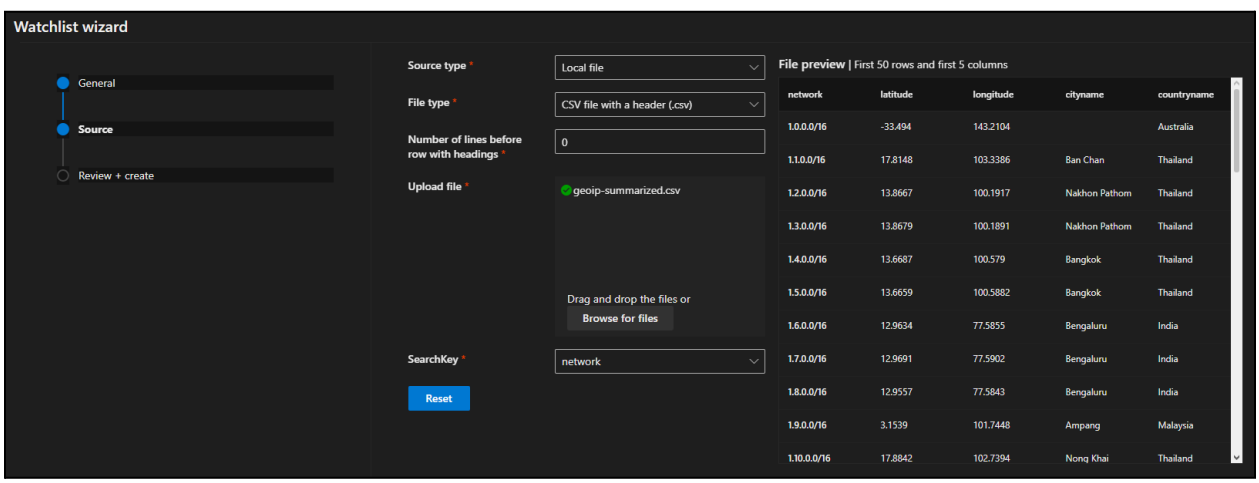
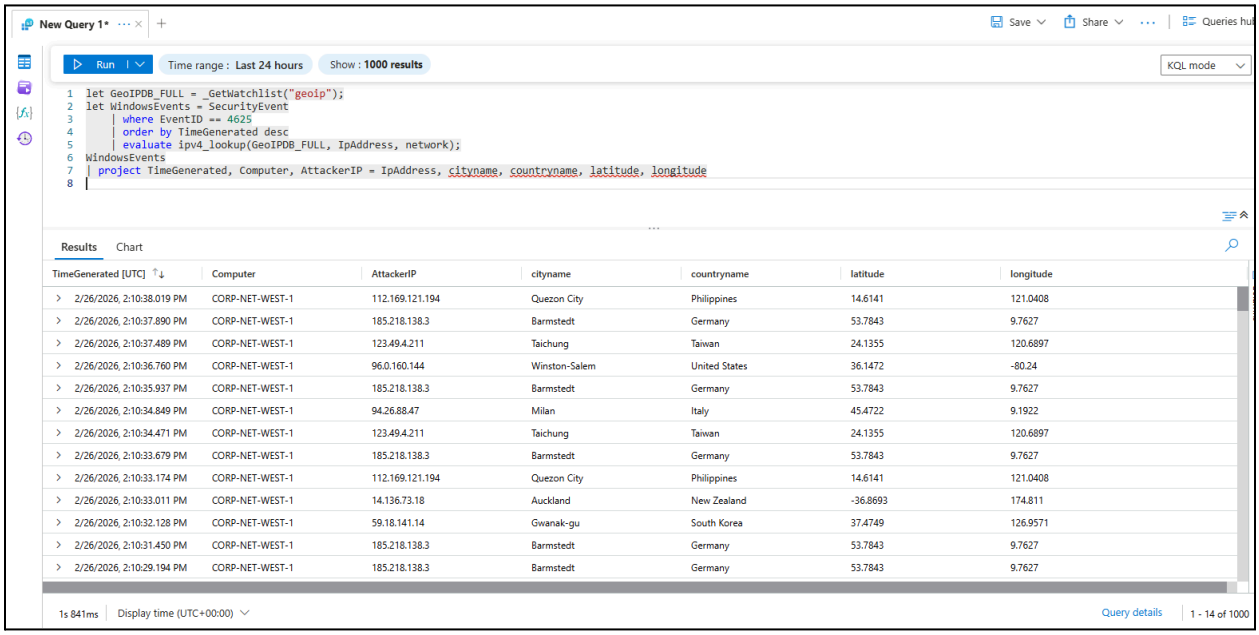


Figure 10.

Observe that the logs now have geographic information, so we can see where the attacks are coming from using these KQL commands:

```
let GeoIPDB_FULL = _GetWatchlist("geoip");
let WindowsEvents = SecurityEvent
    | where EventID == 4625
    | order by TimeGenerated desc
    | evaluate ipv4_lookup(GeoIPDB_FULL, IpAddress, network);
WindowsEvents
| project TimeGenerated, Computer, AttackerIP = IpAddress, cityname,
countryname, latitude, longitude
```



TimeGenerated [UTC]	Computer	AttackerIP	cityname	countryname	latitude	longitude
2/26/2026, 2:10:38.019 PM	CORP-NET-WEST-1	112.169.121.194	Quezon City	Philippines	14.6141	121.0408
2/26/2026, 2:10:37.890 PM	CORP-NET-WEST-1	185.218.138.3	Barmstedt	Germany	53.7843	9.7627
2/26/2026, 2:10:37.489 PM	CORP-NET-WEST-1	123.49.4.211	Taichung	Taiwan	24.1355	120.6897
2/26/2026, 2:10:36.760 PM	CORP-NET-WEST-1	96.0.160.144	Winston-Salem	United States	36.1472	-80.24
2/26/2026, 2:10:35.937 PM	CORP-NET-WEST-1	185.218.138.3	Barmstedt	Germany	53.7843	9.7627
2/26/2026, 2:10:34.849 PM	CORP-NET-WEST-1	94.26.88.47	Milan	Italy	45.4722	9.1922
2/26/2026, 2:10:34.471 PM	CORP-NET-WEST-1	123.49.4.211	Taichung	Taiwan	24.1355	120.6897
2/26/2026, 2:10:33.679 PM	CORP-NET-WEST-1	185.218.138.3	Barmstedt	Germany	53.7843	9.7627
2/26/2026, 2:10:33.174 PM	CORP-NET-WEST-1	112.169.121.194	Quezon City	Philippines	14.6141	121.0408
2/26/2026, 2:10:33.011 PM	CORP-NET-WEST-1	14.136.73.18	Auckland	New Zealand	-36.8693	174.811
2/26/2026, 2:10:32.128 PM	CORP-NET-WEST-1	59.18.141.14	Gwanak-gu	South Korea	37.4749	126.9571
2/26/2026, 2:10:31.450 PM	CORP-NET-WEST-1	185.218.138.3	Barmstedt	Germany	53.7843	9.7627
2/26/2026, 2:10:29.194 PM	CORP-NET-WEST-1	185.218.138.3	Barmstedt	Germany	53.7843	9.7627

Figure 11.

Part 5. Attack Map Creation

Create a new Workbook in Microsoft Sentinel. Delete the prepopulated elements and add a “Query” element, and go to the Advanced Editor tab.

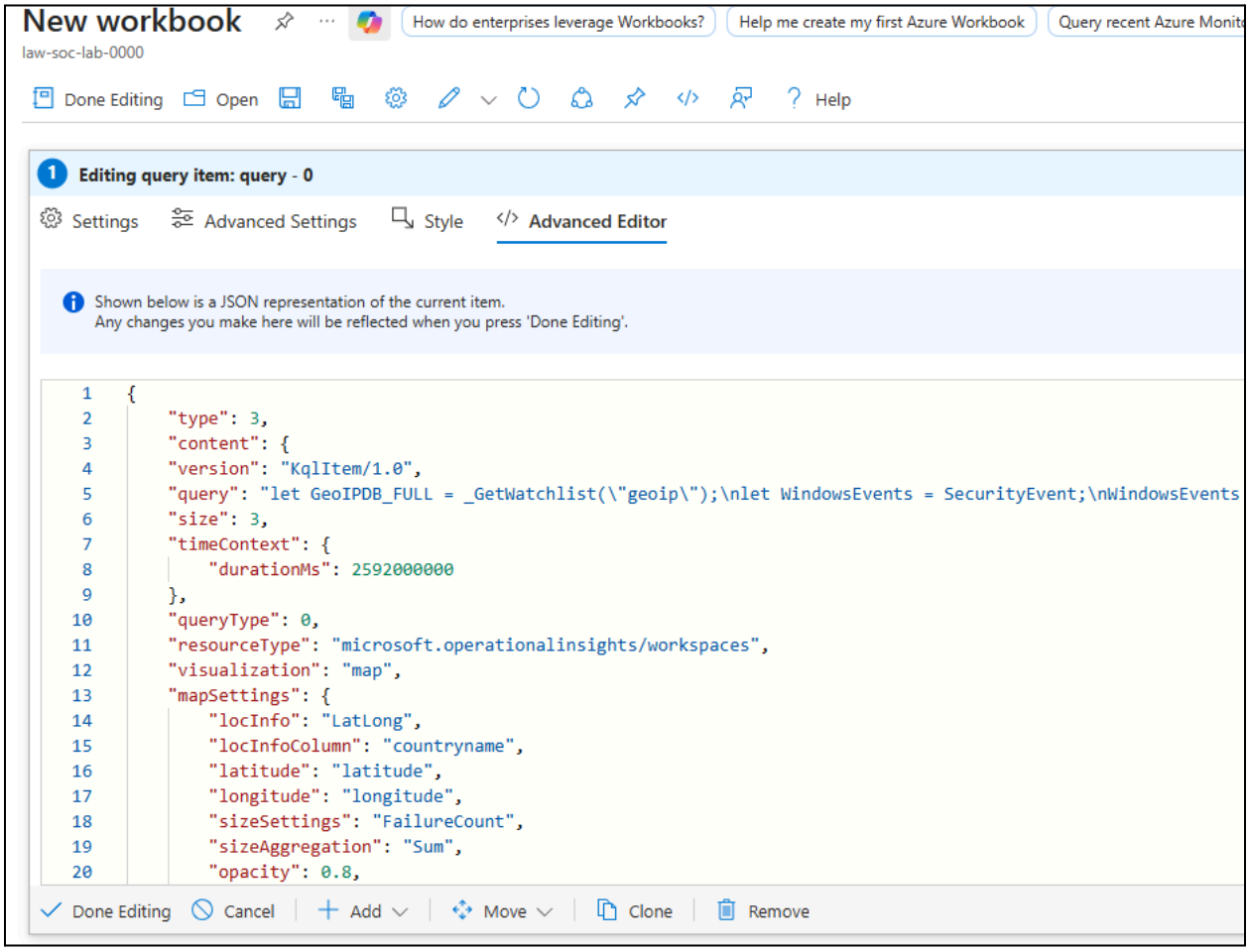


Figure 12.

Paste the JSON Workbook (Attack map): [map.json]


```
{
        "type": 3,
        "content": {
        "version": "KqlItem/1.0",
        "query": "let GeoIPDB_FULL = _GetWatchlist(\"geoip\");\nlet WindowsEvents = SecurityEvent;\nWindowsEvents |
where EventID == 4625\n| order by TimeGenerated desc\n| evaluate ipv4_lookup(GeoIPDB_FULL, IpAddress, network)\n|
summarize FailureCount = count() by IpAddress, latitude, longitude, cityname, countryname\n| project FailureCount,
AttackerIp = IpAddress, latitude, longitude, city = cityname, country = countryname,\nfriendly_location = strcat(cityname, \"
(\", countryname, \")\");",
        "size": 3,
        "timeContext": {
                "durationMs": 2592000000
```

```
        },
        "queryType": 0,
        "resourceType": "microsoft.operationalinsights/workspaces",
        "visualization": "map",
        "mapSettings": {
                "locInfo": "LatLong",
                "locInfoColumn": "countryname",
                "latitude": "latitude",
                "longitude": "longitude",
                "sizeSettings": "FailureCount",
                "sizeAggregation": "Sum",
                "opacity": 0.8,
                "labelSettings": "friendly_location",
                "legendMetric": "FailureCount",
                "legendAggregation": "Sum",
                "itemColorSettings": {
                "nodeColorField": "FailureCount",
                "colorAggregation": "Sum",
                "type": "heatmap",
                "heatmapPalette": "greenRed"
                }
        }
        },
        "name": "query - 0"
}
```


And we will get the attack map.

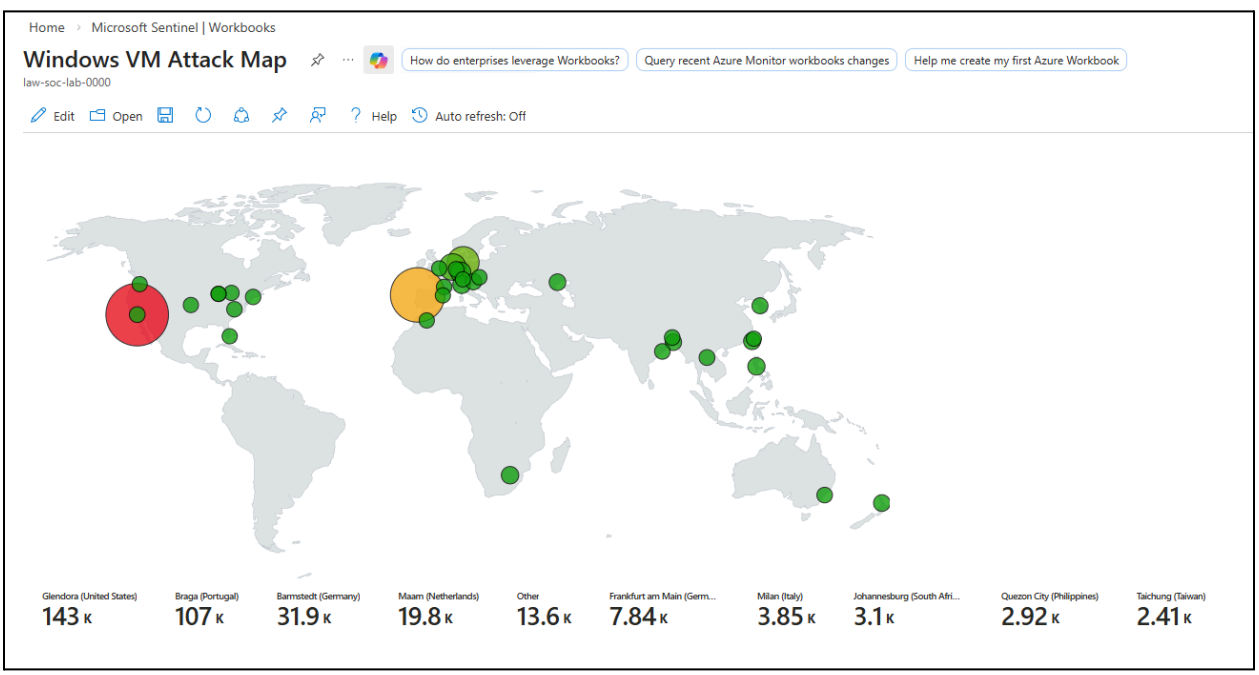


Figure 13.